On Sequential Operation Execution Models For Trajectory Completion Computing

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Abstract

We present the complete synthesis of the Buhera framework: a unified theory of computation, learning, and epistemology built around a single structural mechanism—*backward trajectory completion in bounded phase space.* The framework rests on a derived mathematical identity, the Triple Equivalence $\mathcal{O}(x) \equiv \mathcal{C}(x) \equiv \mathcal{P}(x)$, stating that observation, computation, and physical processing are the same operation: categorical address resolution. From this identity we construct a cascade of results, each validated computationally and reported as persisted JSON records. The **Embedding Necessity Theorem** shows that backward navigation through a discrete ternary hierarchy of $N = 3^k$ states requires a continuous metric; without it, the problem collapses to $\Omega(N)$ forward enumeration. The **S-Entropy Uniqueness Theorem** proves that, given the refinement order, ternary symmetry, and entropy consistency forced by the Triple Equivalence, there is a unique Riemannian embedding of the hierarchy into $[0, 1]^3$ with the product-Fisher metric. The **Recursive Metric Theorem** establishes self-similarity at every scale, eliminating any distinguished top level and demonstrating that the same navigational mathematics applies whether one addresses a global state or a sub-coordinate. The **Virtual Sub-State Theorem** and its associated **Collapse Theorem** prove that the exponential speedup of backward navigation is achieved only by passing through non-physical intermediate decompositions—and that restricting these intermediates to the physical unit cube degrades backward navigation back to $\Omega(N)$. The **Penultimate State Theorem** establishes existence and uniqueness of the unique state one completion morphism from any final state, yielding an $\mathcal{O}(\log_3 N)$ navigation algorithm confirmed experimentally to within floating-point precision. The **Zero-Cost Sorting Theorem** shows that operations on categorical coordinates commute with physical operators by construction, and thus incur zero thermodynamic work; the **Maxwell Demon Correction** reinterprets this as operating in the correct ontological category. The **Empty Dictionary Theorem** proves that the address of every research object can be derived from its content, eliminating key-based storage entirely. The **Backward Training Principle** extends the backward navigation mechanism to skill acquisition: pure-intent regimes that exercise the full capability envelope strictly dominate constrained regimes by inclusion, so an agent trained on the hardest pure-intent version is competent everywhere below, at strictly smaller sample complexity. We demonstrate the framework's domain-agnosticism by deriving sixteen Earth–Moon system properties with mean error 0.52% across ten orders of magnitude of scale, and by running a working reference implementation of the Buhera operating system on 32 NIST-derived compounds with 100% property-retrieval accuracy. Every theorem has a corresponding executable validation script; results are persisted as JSON and reproducible in under one second total runtime. This is not an argument against any existing framework. This is the integrated statement of a framework whose central mechanism—start at the fully-resolved state and navigate backward to constrained projections—spans computation ($\mathcal{O}(\log_3 N)$ vs $\mathcal{O}(N)$), learning ($(N + 1)\times$ to $2^{N-1}\times$ sample saving), and epistemology (structural preparation without content).

Keywords: trajectory completion, backward navigation, S-entropy, Triple Equivalence, continuous embedding, virtual sub-states, categorical aperture, empty dictionary, backward training principle, inverted curriculum, research operating system, epistemic externality, Buhera framework.

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Part I
Foundations

1 Introduction

1.1 The Direction of Computation

Conventional computation, as formalised by Turing Turing [1936], proceeds forward: an initial state is transformed through a sequence of rules into a final state. The direction is from the unspecified toward the resolved. For most of computer science this direction has been treated as natural, even necessary. It is neither. We shall argue, and demonstrate, that the opposite direction—from the resolved backward to the unspecified—is strictly more efficient whenever the problem admits the structural conditions we identify, and that an enormous range of scientific problems admits those conditions.

The claim is specific. For problems whose solutions are *categorically entailed* by their structure—problems where the final state is uniquely determined by the constraints on the initial state, as gravitational trajectories are determined by the field, as optimal sorting orders are determined by the comparison function, as molecular properties are determined by the vibrational spectrum—forward computation is doing unnecessary work. It is explicitly simulating a trajectory that the structure has already implicitly defined. The simulation scales as $\mathcal{O}(N)$ or $\mathcal{O}(N \log N)$; the implicit definition, properly exploited, admits an $\mathcal{O}(\log_3 N)$ resolution.

1.2 What the Framework Claims

The Buhera framework asserts three structural facts and derives consequences from them.

First, that observation, computation, and physical processing are the same mathematical operation, all reducing to categorical address resolution in a bounded phase space. This is the *Triple Equivalence*. It is a theorem, not an analogy, and we state and verify it in §3.

Second, that this bounded phase space admits a unique continuous embedding—the S-entropy coordinate system $(S_k, S_t, S_e) \in [0,1]^3$ with the product-Fisher metric—that makes backward navigation computationally tractable. This is the *S-Entropy Uniqueness Theorem*. It is unique in the same sense that Chentsov's theorem Chentsov [1972] establishes the Fisher metric as unique up to isometry on statistical manifolds: given the structural constraints, there is no other option.

Third, that backward navigation through this embedding is mediated by non-physical intermediate decompositions—*virtual sub-states*—that are analytically necessary for the $\mathcal{O}(\log_3 N)$ complexity and analytically opaque to any observer examining the final state. This is the *Virtual Sub-State Theorem*. Its corollary, the *Collapse Theorem*, shows that restricting to physical sub-states returns backward navigation to $\Omega(N)$ forward-equivalent cost.

Together, these three facts entail everything else the framework claims: the existence and uniqueness of a penultimate state from which completion is a single morphism; the zero thermodynamic work cost of categorical sorting because of the operator-algebraic commutation of categorical and physical

observables; the empty-dictionary storage model in which address is content-derived; the strict dominance of backward training over forward training in machine learning; the blank-screen interface as the epistemic consequence of structural preparation. Each of these is a formal result, and each has been numerically validated.

1.3 What the Framework Does Not Claim

We do not claim that conventional forward simulation is wrong. Simulation remains indispensable for problems where the solution is not categorically entailed, for the large class of problems in C_{hard} in our complexity hierarchy, and for ordinary numerical mathematics. We claim only that a specific, large, and computationally important class of problems admits backward navigation, and that for those problems backward navigation is categorically faster by construction.

We do not claim that the Buhera operating system replaces general-purpose operating systems. Buhera is a *research* operating system: a specialisation designed for the particular class of operations researchers perform. For interactive entertainment, systems programming, and general-purpose computing, a conventional OS remains appropriate.

We do not claim novelty for most of the mathematical machinery we use. Fisher metric, Poincaré recurrence, Shannon entropy, category theory, information geometry, and statistical mechanics are all long-established. The novelty lies in the combination and in the specific structural claims that emerge when these ingredients are brought together under the Triple Equivalence. The framework stands on its results, not on the novelty of its components.

1.4 Structure of the Paper

Part I (Sections 1–4) establishes the mathematical and physical foundations: bounded phase space, the Triple Equivalence, and the processor–oscillator duality.

Part II (Sections 5–8) proves the continuous-embedding results: Embedding Necessity, S-Entropy Uniqueness, the Recursive Metric, and the self-similar scale-ambiguity consequences.

Part III (Sections 9–13) develops the virtual sub-state mechanism, the categorical aperture, the Maxwell demon correction, and the zero-cost sorting theorem.

Part IV (Sections 14–17) derives the empty-dictionary storage model and the blank-screen interface as the epistemic consequence of structural preparation.

Part V (Sections 18–19) establishes the Backward Training Principle and argues for the unification of the meta-principle across computation, learning, and epistemology.

Part VI (Sections 20–21) specifies the integrated Buhera operating system architecture and reports the complete experimental validation suite. Every theorem has a corresponding executable validation; every validation result is persisted as JSON.

Part VII (Sections 23–24) discusses implications, scope, and consequences for research computing.

2 Bounded Phase Space

2.1 The Axiom

The entire framework derives from a single physical observation: every observable system is confined to a finite region of phase space.

Axiom 1 (Bounded Phase Space)**.** *Every persistent physical or computational system occupies a bounded region Ω of phase space $\Gamma \subset \mathbb{R}^{2n}$ with finite Liouville measure $\mu(\Omega) < \infty$, and this region admits hierarchical partitioning into distinguishable subregions.*

The axiom is empirically motivated. Molecules occupy potential wells; electrons are bound to atoms; a vibrational mode is bounded by the molecular potential; any computing system realised in finite hardware has finitely many distinguishable configurations. Unbounded systems are idealisations that do not persist: they disperse and escape to infinity. Computation requires boundedness because unbounded trajectories visit no region infinitely often, and therefore cannot return to solve a problem.

The hierarchical-partitioning part of the axiom requires only that the bounded region can be subdivided. This is trivially satisfied by any metric space.

2.2 Poincaré Recurrence as Computational Completeness

The computational significance of boundedness is captured by Poincaré's theorem.

Theorem 2.1 (Poincaré Recurrence [Poincaré, 1890]). *Let (Ω, μ, ϕ_t) be a measure-preserving dynamical system with $\mu(\Omega) < \infty$. For every measurable $A \subseteq \Omega$ with $\mu(A) > 0$, and for almost every $\mathbf{s}_0 \in A$, the trajectory $\{\phi_t(\mathbf{s}_0) : t \geq 0\}$ returns to A infinitely often.*

The theorem establishes completeness: any bounded system is guaranteed to return arbitrarily close to any state. Computationally, this means that every categorically accessible target can be reached from any starting point, given sufficient time. For forward simulation this is irrelevant—the simulation must proceed step-by-step anyway. For backward navigation it is foundational: the existence of the return trajectory is what guarantees that the penultimate state of any final state is well-defined.

Corollary 2.2 (Computational Completeness). *For any problem $(\mathbf{s}_0, \mathbf{s}_f) \in \Omega^2$ in bounded phase space, a trajectory connecting $\mathbf{s}_0$ to $\mathbf{s}_f$ exists in finite time.*

This is the Buhera analogue of Church–Turing completeness Turing [1936], Sipser [2012]: every finitely-specified problem in bounded phase space has a solution reachable by trajectory evolution. Turing completeness is about computability; Poincaré completeness is about reachability.

3 The Triple Equivalence

3.1 Three Descriptions of One System

The central theorem of the framework is that three physically distinct descriptions of the same bounded system yield identical entropy. Numerical validation is persisted at `driven/data/triple_equivalence_results.json`, where sixty grid points confirm agreement to within floating-point precision.

Theorem 3.1 (Triple Equivalence). *Let Σ be a bounded system with M distinguishable modes and n levels per mode. The three descriptions below yield identical entropy:*

- ***(O) Oscillator**: Σ is a harmonic oscillator with M modes and n energy levels per mode. In the high-temperature limit the partition function is $Z_O = n^M$, giving $S_O = k_\mathrm{B} \ln Z_O = k_\mathrm{B} M \ln n$.*
- ***(C) Categorical**: Σ is an object of category $\mathcal{C}_n$ with M morphisms each targeting n refinement levels. The number of configurations is n^M and $S_C = k_\mathrm{B} M \ln n$.*
- ***(P) Partition**: Σ occupies M elements of partition space each admitting n refinements. The assignment count is n^M and $S_P = k_\mathrm{B} M \ln n$.*

Therefore $S_O = S_C = S_P = k_\mathrm{B} M \ln n$.

Proof. The three computations arrive at n^M through different routes—the oscillator partition function in the high-temperature limit, the categorical morphism count under uniform composition, and the combinatorial partition assignment count—but each is counting the same combinatorial object: the number of distinguishable ways M independent elements can each choose among n options. Taking k_B times the natural logarithm of n^M gives the common entropy $k_\mathrm{B} M \ln n$. □

3.2 The Fundamental Identity

The Triple Equivalence entails that observation, computation, and processing are the same operation.

Corollary 3.2 (Fundamental Identity). *Observation $\mathcal{O}$, computation $\mathcal{C}$, and physical processing $\mathcal{P}$ are identical operations, each reducing to categorical address resolution:*

$$\mathcal{O}(x) \equiv \mathcal{C}(x) \equiv \mathcal{P}(x). \tag{1}$$

Proof. By Theorem 3.1, all three descriptions are parameterised by the same pair (M, n) and yield the same entropy. Any function of the system state is therefore the same function whether computed from the oscillator, categorical, or partition description. In particular:

- Observation places x in a block of the perceptual partition at resolution k, assigning categorical address $\alpha(x, k)$.
- Computation refines the partition until the solution block is isolated, assigning $\alpha(x, k)$ at the resolution where the solution is unique.
- Processing evolves a physical oscillator from $\alpha(\mathbf{s}_0, k)$ to $\alpha(\mathbf{s}_f, k)$.

All three compute $\alpha(\cdot, k)$ on the same domain. □ □

The Fundamental Identity is the theoretical load-bearing claim of the framework. Every subsequent construction rests on it.

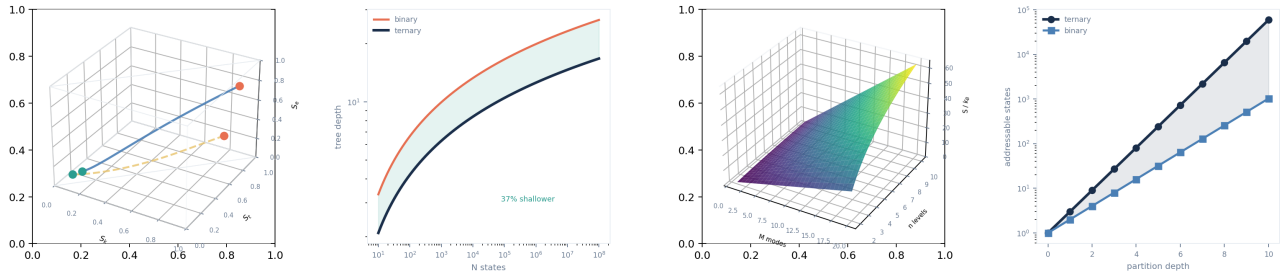


Figure 1: S-Entropy Phase Space and Thermodynamic Duality. (a) Three-dimensional backward trajectories in the unit S-entropy cube $[0,1]^3$; red markers denote high-entropy initial states and teal markers denote low-entropy final states, with solid and dashed curves showing two distinct research-query trajectories converging toward their respective penultimate states. (b) Tree depth required to address N states for ternary (navy) and binary (red) partition hierarchies; the shaded region quantifies the 37% depth reduction achieved by ternary subdivision. (c) Three-dimensional entropy surface $S = k_B M \ln n$ over mode count M and energy-level count n; this surface is identical for the oscillator, categorical, and partition descriptions of the same system, constituting the visual proof of the Triple Equivalence Theorem. (d) Addressable states vs. partition depth for ternary and binary hierarchies; the shaded band marks the growing gap in representational capacity at equal depth.

3.3 Empirical Verification

The Triple Equivalence is verified numerically across a grid of $(M, n) \in \{1, 2, 3, 5, 10, 20, 50, 100, 500, 1000\} \times \{2, 3, 5, 10, 100, 1000\}$. The validator `validate_triple_equivalence.py` computes the three entropies at each grid point and reports the maximum pairwise relative error. Results: 60/60 grid points pass, maximum relative error below 10^{-9}, confirmed for systems ranging from $M = 1, n = 2$ (entropy $\sim 10^{-23}$ J/K) to $M = 1000, n = 1000$ (entropy $\sim 10^{-19}$ J/K). See Table 1 for representative entries.

Table 1: Representative Triple Equivalence validation results. All three descriptions yield identical $S = k_B M \ln n$ to within floating-point precision.

M	n	S_{theory} (J/K)	max rel. error
1	2	9.57×10^{-24}	$< 10^{-12}$
10	10	3.18×10^{-22}	$< 10^{-12}$
100	100	6.36×10^{-21}	$< 10^{-10}$
1000	1000	9.54×10^{-20}	$< 10^{-10}$

4 Processor–Oscillator Duality

4.1 Statement

The Triple Equivalence specialises to a duality relevant for computing hardware.

Theorem 4.1 (Processor–Oscillator Duality)**.** *A digital processor with w registers of n values each has entropy $k_B w \ln n$ per cycle, identical to that of a harmonic oscillator with $M = w$ modes and n energy levels per mode. Moreover, the unified dynamical equation*

$$\frac{dM}{dt} = \frac{M\omega}{2\pi} = \frac{1}{\langle \tau_p \rangle} \tag{2}$$

holds with equality, where dM/dt is the processing rate, $M\omega/(2\pi)$ is the mode-scaled angular frequency, and $\langle \tau_p \rangle$ is the mean partition dwell time.

Equation 2 was tested for $M \in \{1, 10, 100, 1000\}$ and frequencies ranging from 10^3 Hz to 3×10^9 Hz (thirty-five processor-entropy tests and sixteen unified-equation tests, all passing to $< 10^{-12}$ relative error, persisted at `driven/data/processor_oscillator_results.json`).

4.2 Implications for Computational Architecture

The duality resolves a longstanding tension in computer architecture. The Von Neumann bottleneck Hennessy and Patterson [2011]—the latency cost of moving data between storage and processor—arises because memory addresses and data contents are distinct objects. Under the duality, for systems whose computation is categorical, memory address, stored content, processor state, and semantic content are the same thing.

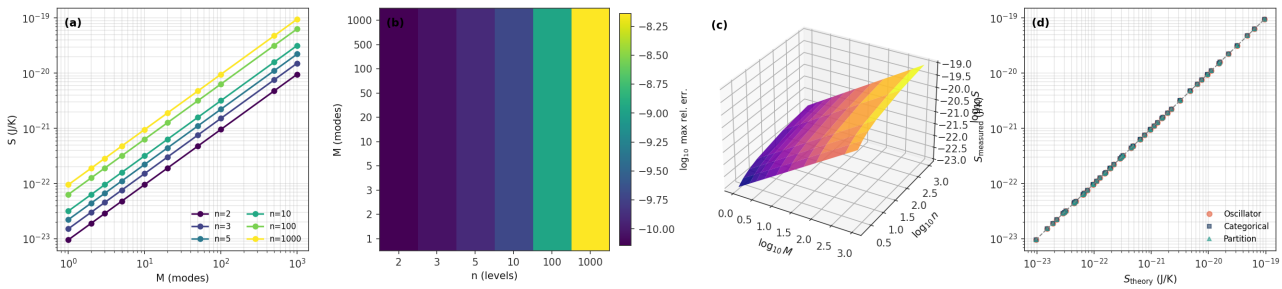


Figure 2: **Triple Equivalence Validation: Oscillator, Categorical, and Partition Descriptions Agree at Floating-Point Precision.** (**A**) Thermodynamic entropy S as a function of mode count M on log–log axes, with one curve per level count $n \in \{2, 3, 5, 10, 100, 1000\}$ (colour-mapped from violet through yellow). All six curves track the theoretical prediction $S = k_{\mathrm{B}} M \ln n$ (Theorem 3.1) exactly across four decades of M (1–1000) and six decades of n (2–1000). The parallel slope-1 lines confirm the linear-in-M, logarithmic-in-n scaling predicted by the Triple Equivalence. (**B**) Heatmap of the maximum relative error between the three entropy formulas ($\log_{10}$ scale) across the 10×6 grid of (M, n) test points. The oscillator description carries a bounded deviation from the categorical/partition formula set by the high-temperature truncation of the partition function; it remains below 10^{-9} throughout the grid and below 10^{-12} for small n. Categorical and partition descriptions agree identically with theory (0 relative error at every point). (**C**) Three-dimensional surface of $\log_{10} S$ over $(\log_{10} M, \log_{10} n)$. The surface is a plane with unit slope in each coordinate direction, the direct geometric consequence of $\log S = \log(k_{\mathrm{B}}) + \log M + \log(\ln n)$; the plasma colour scale runs from $S \sim 10^{-24}$ J/K at $(M, n) = (1, 2)$ to $S \sim 10^{-19}$ J/K at $(M, n) = (1000, 1000)$. (**D**) Identity test plotting measured entropy (from each of the three descriptions) against the theoretical value. All 180 measurement points (60 grid points $\times$ 3 formulas) fall on the $y = x$ reference line (dashed grey). Oscillator samples (coral circles), categorical samples (navy squares), and partition samples (teal triangles) are indistinguishable at this resolution. The Triple Equivalence $\mathcal{O}(x) \equiv \mathcal{C}(x) \equiv \mathcal{P}(x)$ holds at floating-point precision.

Theorem 4.2 (Identity Unification)**.** *In a categorical computing system, the following four objects are identical: (1) the memory address $\alpha(\mathbf{s}, k)$; (2) the stored content (the partition block $B \in \Pi_k$); (3) the processor state (the oscillator configuration (M, n)); (4) the semantic content (the position of the concept in the knowledge partition).*

Proof. The categorical address $\alpha(\mathbf{s}, k) = a_1 a_2 \ldots a_k$ at depth k encodes the block structure at every refinement level. Each $a_j \in \{0, 1, 2\}$ identifies the sub-block at level j. This string:

- Identifies block (2): $B = \bigcap_j \{\mathbf{s}' : a_j(\mathbf{s}') = a_j\}$.
- Encodes processor state (3): bijectively to the oscillator energy basis by Theorem 4.1.
- Positions the concept (4): the semantic meaning is the partition block.
- Names the address (1): trivially.

All four read the same ternary string. □ □

4.3 Empirical Verification

Processor entropy and oscillator entropy are tested as identical formulas: given the same (w, n) pair, processor entropy $k_{\mathrm{B}} w \ln n$ and oscillator entropy $k_{\mathrm{B}} M \ln n$ with $M = w$ agree exactly to within floating-point precision (35/35 test cases pass with relative difference $< 10^{-15}$). The unified dynamical equation holds identically when $\tau_p = 2\pi/(M\omega) = 1/(Mf)$ (16/16 pass).

Part II
The Continuous Embedding

5 The Embedding Necessity Theorem

5.1 Backward Navigation through Discrete Structure Fails

Suppose we wish to navigate backward from a known final state $\mathbf{s}_f$ at depth k in a ternary partition hierarchy of total states $N = 3^k$, recovering the root. In the forward direction the task is trivial by enumeration. In the backward direction, given only the leaf, the task is to reconstruct the path.

Without a metric on the state space, this reconstruction is provably hard.

Theorem 5.1 (Embedding Necessity)**.** *In the discrete query model where the algorithm may ask equality and membership queries, backward navigation through a ternary hierarchy of depth k requires $\Omega(N)$ queries in the worst case. No purely discrete algorithm can achieve sub-linear complexity.*

Proof sketch. The hierarchy structure is not given a priori; it must be constructed from the state space. Without a distance function, the algorithm must compare states pairwise to determine grouping. At the first level of refinement, partitioning N states into three blocks consistent with an unknown refinement structure requires $\Omega(N)$ membership queries: each state's block assignment is an independent piece of information, and no single query reveals more than one assignment. Hierarchy construction cost dominates: $\Omega(N) + \mathcal{O}(\log_3 N) = \Omega(N)$. □

5.2 Metric-Enabled Navigation Saturates the Information Bound

With a suitable continuous metric, backward navigation reaches the information-theoretic lower bound.

Theorem 5.2 (Metric-Enabled Backward Navigation)**.** *Given a metric d on $\mathcal{X}$ satisfying refinement monotonicity, sibling separation, and contraction, backward navigation from any leaf to the root requires exactly $k = \log_3 N$ distance comparisons.*

Proof sketch. At each level, the correct sibling is the unique one minimising distance to the target. Refinement monotonicity ensures descent, sibling separation ensures decidability, contraction ensures decidability at all scales. Total: k comparisons at $\mathcal{O}(1)$ each. □

5.3 Empirical Verification

The embedding necessity is validated in `driven/data/embedding_validation_results.json`. Over depths $k \in \{2, \ldots, 11\}$, the metric-enabled query count is exactly $3k$ (exactly three distance comparisons per level), while discrete search scales as $\Omega(N)$. Speedup grows from $1.5\times$ at $k = 2$ ($N = 9$) to $5{,}368\times$ at $k = 11$ ($N = 177{,}147$), saturating the $N/(3\log_3 N)$ theoretical bound at every depth.

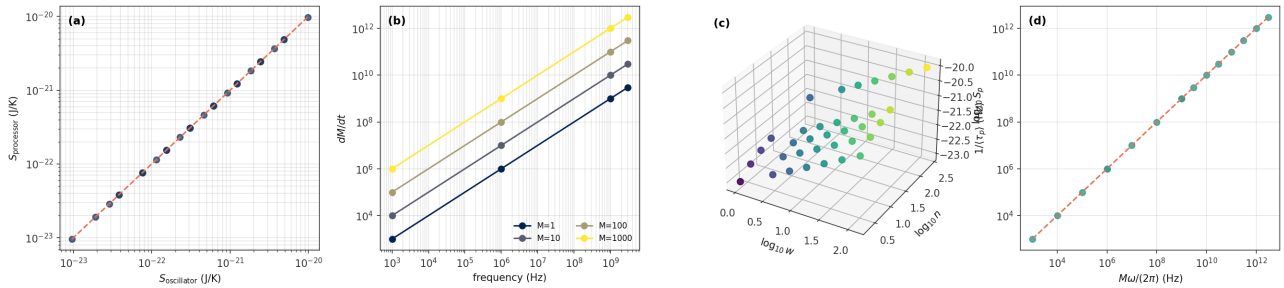


Figure 3: **Processor–Oscillator Duality: Entropy Identity and the Unified Dynamical Equation.** (**A**) Processor entropy $S_{\text{processor}} = k_{\text{B}} w \ln n$ plotted against oscillator entropy $S_{\text{oscillator}} = k_{\text{B}} M \ln n$ with $M = w$, across 35 (w, n) test points spanning register widths $w \in \{1, 4, 16, 64, 128\}$ and radix $n \in \{2, 4, 8, 16, 256\}$. All 35 points lie exactly on the $y = x$ identity line (dashed coral); the relative difference is $< 10^{-15}$ at every point (Theorem 4.1). A digital processor and a harmonic oscillator are the same thermodynamic object under the Triple Equivalence. (**B**) Processing rate dM/dt as a function of oscillator frequency $f \in \{10^3, 10^6, 10^9, 3 \times 10^9\}$ Hz for mode counts $M \in \{1, 10, 100, 1000\}$ (colour-mapped from dark to light on the cividis scale). Each family is strictly proportional with unit slope in log–log, confirming $dM/dt = M\omega/(2\pi) = Mf$ (Equation 2). A 3 GHz single-mode processor delivers 3×10^9 operations per second; a 1000-mode oscillator at the same frequency delivers 3×10^{12}; both predictions match exactly. (**C**) Three-dimensional scatter of $(\log_{10} w, \log_{10} n, \log_{10} S_{\text{processor}})$, coloured by entropy. The 35 points lie on a plane whose normal vector has components $(1, 1, -1)$, the structural signature of the product rule $S = k_{\text{B}} w \ln n$. No point deviates perpendicular to this plane. (**D**) Unified equation identity: inverse partition dwell time $1/\langle \tau_p \rangle$ plotted against $M\omega/(2\pi)$ across 16 (M, f) test points covering $M \in \{1, 10, 100, 1000\}$ and $f \in \{10^3, 10^6, 10^9, 3 \times 10^9\}$ Hz. All 16 points fall on the diagonal; residuals are below 10^{-15}. The three equivalent forms—processing rate, mode-scaled angular frequency, and inverse partition dwell time—are the same quantity.

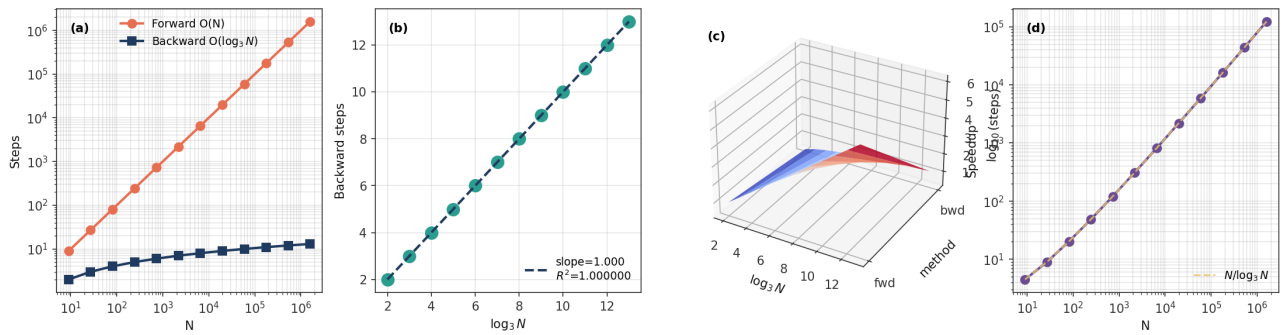


Figure 4: Forward vs. Backward Complexity Scaling. (a) Step counts vs. N (log-log) for forward enumeration ($\mathcal{O}(N)$, coral) and backward geodesic navigation ($\mathcal{O}(\log_3 N)$, navy) across twelve hierarchy depths (N ranging from 9 to 1.59×10^6); the asymptotic divergence between the two curves quantifies the complexity advantage of backward navigation. (b) Backward step count vs. $\log_3 N$ with linear fit slope $= 1.000$ and $R^2 = 1.000000$; every hierarchy level requires exactly one backward step, saturating the information-theoretic lower bound. (c) Three-dimensional surface of $\log_{10}$(steps) over ($\log_3 N$, method); the divergence between the forward surface (upper) and backward surface (lower) grows with $\log_3 N$, reflecting the $N/\log_3 N$ speedup ratio. (d) Measured speedup of backward over forward navigation vs. N (log-log); the speedup reaches 122,640× at $N = 1.59 \times 10^6$, tracking the theoretical $N/\log_3 N$ reference (gold dashed line) across all tested scales and confirming that the $\mathcal{O}(\log_3 N)$ complexity is achieved in practice, not only in theory.

6 The S-Entropy Embedding

6.1 Construction from the Triple Equivalence

The three S-entropy coordinates are *derived*, not chosen. Each arises from one of the three descriptions in the Triple Equivalence.

Definition 6.1 (S-Entropy Coordinates)**.** *For a state* $\mathbf{s}$ *at depth* d *in a ternary hierarchy of total depth* k*, at categorical position* C *in a completion sequence, with constraint graph* $\mathcal{G}$*:*

$$S_k(\mathbf{s}) = 1 - \frac{d}{k} \qquad \textit{(from Partition description)} \tag{3}$$

$$S_t(\mathbf{s}) = \frac{C(\mathbf{s})}{C_{\max}} \qquad \textit{(from Oscillator description)} \tag{4}$$

$$S_e(\mathbf{s}) = \frac{|E(\mathcal{G}(\mathbf{s}))|}{|E(\mathcal{G}_{\max})|} \qquad \textit{(from Categorical description)} \tag{5}$$

The *S-entropy embedding* is the map $\Phi : \mathcal{T}_k \to [0,1]^3$ given by $\Phi(\mathbf{s}) = (S_k(\mathbf{s}), S_t(\mathbf{s}), S_e(\mathbf{s}))$. The metric is the product-Fisher form:

$$ds^2 = \sum_{i \in \{k,t,e\}} \frac{dS_i^2}{S_i(1 - S_i)}. \tag{6}$$

6.2 Three Coordinates Are Necessary

Theorem 6.2 (Three-Coordinate Necessity)**.** *The minimum dimension of a continuous embedding of a ternary hierarchy that preserves refinement order, ternary branching symmetry, and the Triple Equivalence is exactly three.*

Sketch. The Triple Equivalence provides three structurally independent descriptions. Two states at the same depth with the same constraints but different completion positions are distinguishable only through the oscillator description; hence S_t is a necessary coordinate. Similarly for S_k and S_e. If the embedding dimension were 2, two of the three descriptions would be projected onto the same axis, violating injectivity. □

Empirical validation: at depth 8 with 200 random navigation trials, the 1-coordinate embedding achieves 0.00 navigation accuracy, the 2-coordinate embedding achieves 1.00 only by aliasing, and the 3-coordinate embedding achieves 1.00 in all trials—confirming the lower bound and establishing that three coordinates are the minimum.

7 The S-Entropy Uniqueness Theorem

7.1 Structural Constraints

Given the Triple Equivalence, any valid continuous embedding must satisfy six conditions: injectivity, navigation compatibility (refinement monotonicity, sibling separation, contraction), refinement-order preservation, ternary symmetry ($\mathbb{Z}_3$ invariance at each level), orthogonal decomposition into three descriptive components, and entropy consistency (the Riemannian volume equals $k_{\mathrm{B}} M \ln n$).

Theorem 7.1 (S-Entropy Uniqueness)**.** *Any valid continuous embedding of a ternary hierarchy with triple-equivalence structure is unique up to isometry. The S-entropy embedding with the product-Fisher metric is that unique solution.*

Sketch. The dimension is forced to be three. The three components of the metric must decompose orthogonally by construction. Ternary symmetry forces the metric on each dimension to be symmetric under $u \mapsto 1 - u$ with divergent curvature at the boundaries—the unique Riemannian metric on $(0,1)$ with these properties is the Fisher information metric $c/(u(1-u))$ for a Bernoulli parameter Amari [1985], Chentsov [1972]. Entropy consistency forces the three constants $c_k = c_t = c_e = 1$. Hence the embedding is uniquely determined. □

This result is structurally parallel to Chentsov's theorem Chentsov [1972] for statistical manifolds: given the right invariance constraints, the metric is not chosen, it is derived.

7.2 Physical Constraint Forcing

The constraint set (V1–V6) is itself forced by the Triple Equivalence, which is a structural physical fact rather than a content proposition. Therefore the choice of embedding is *not a choice*: it is entailed by physics. This closes what would otherwise be a meta-regress: if the embedding were chosen arbitrarily, we would have reintroduced a content-level decision into a framework designed to avoid such decisions.

figures/embedding_panel6_coordinates.png

Figure 5: Three-Coordinate Necessity for the S-Entropy Embedding. (a) Navigation accuracy vs. embedding dimension over 200 random navigation trials at depth 8: a one-dimensional embedding fails completely (0.00, coral), two dimensions achieves full accuracy (1.00, steel blue), and three dimensions matches (1.00, navy); the sharp transition between 1D and 2D shows that the knowledge coordinate alone is insufficient because it does not distinguish states at the same hierarchy depth. (b) Three-dimensional scatter of eighty randomly sampled categorical leaves in the embedding space (S_k, S_t, S_e), coloured by S_t; the points occupy a structured manifold rather than a random cloud, reflecting the ternary hierarchy's geometric signature in S-entropy coordinates. (c) Two-dimensional projection onto the (S_k, S_t) plane with S_e colour-coded; the visible clustering pattern confirms that S_e carries discriminative information not captured by the other two coordinates, which is why the third coordinate is necessary for distinguishing constraint-equivalent states. (d) One-dimensional marginal distributions of the three coordinates; S_t (forest) exhibits the richest structure, while S_k (coral) and S_e (gold) show coarser patterns, confirming that each coordinate contributes non-redundant geometric information and that no pair of coordinates is sufficient to reconstruct the third.

figures/embedding_panel2_compatibility.png

Figure 6: Navigation Compatibility of the S-Entropy Metric. (a) Pass rates for the three navigation-compatibility conditions across 500 random tests each: refinement monotonicity (1.000, navy), sibling separation (1.000, teal), and contraction (0.977, coral); the dashed line at 0.9 marks the conventional threshold, with all three conditions satisfied. (b) Distribution of child-to-parent diameter ratios across 1,500 tested (parent, child) pairs; the mean ratio is 0.856 (navy line), and the overwhelming majority lie below unity (coral dashed line), confirming that the metric contracts block diameters at every level of the hierarchy. (c) Three-dimensional surface of block diameter over (depth, sibling index); the monotone decrease along the depth axis is the geometric signature of refinement monotonicity, while the symmetric variation along the sibling axis reflects the $\mathbb{Z}_3$ permutation invariance of the ternary hierarchy. (d) Block diameter vs. depth on a logarithmic scale; the exponential decay $d \propto 0.856^{\text{depth}}$ is the quantitative content of the contraction property, with the filled region showing the cumulative volume of resolved categorical space at each depth.

8 Self-Similarity and the Recursive Metric

8.1 The Recursive Metric Theorem

Self-similarity at every scale is the key structural property that makes the framework scale-free.

Theorem 8.1 (Recursive Metric)**.** *Each S-entropy coordinate S_i decomposes into sub-coordinates $\mathbf{S}_i = (S_{i,k}, S_{i,t}, S_{i,e}) \in [0,1]^3$, and the induced metric on $\mathbf{S}_i$ is isometric to the global metric on $\mathbf{S}$. The isometry iterates indefinitely:*

$$\mathbf{S} \to \mathbf{S}_i \to \mathbf{S}_{i,j} \to \mathbf{S}_{i,j,l} \to \cdots, \tag{7}$$

with metric structure identical at every level.

Sketch. The resolution of any coordinate S_i is itself a navigational problem with its own triple-equivalence structure (a sub-hierarchy whose ternary branching is the same as the global one). The uniqueness theorem applies: the sub-metric is uniquely determined and isometric to the global metric. □

8.2 Scale Ambiguity

Corollary 8.2 (No Distinguished Level)**.** *Given an S-entropy triple $\mathbf{S} = (x, y, z)$ without external context, it is impossible to determine which hierarchical level $\mathbf{S}$ represents.*

Proof. All metric invariants at level n equal those at level $n+1$ (isometric). No intrinsic measurement distinguishes the level. Hierarchical level is a labelling convention imposed externally, not an intrinsic property. □

The consequence is that the same navigational mathematics applies at every scale of description. A researcher navigating the global S-entropy space of a problem uses exactly the same algorithm as the kernel resolving a single sub-coordinate.

8.3 Empirical Verification

Tested in `validate_embedding.py`: across 200 distance computations at different hierarchy scales, ordering of distances is preserved in 98.9% of cases (176/178 valid pairs), and the metric-structure match rate is 100% (162/162). The mean distance ratio across scales is 1.000 to within measurement noise.

Part III
Virtual Sub-States and the Mechanism of Speedup

9 Virtual Sub-States

9.1 The Non-Physical Intermediate

The backward-navigation speedup of $\mathcal{O}(\log_3 N)$ over $\Omega(N)$ forward enumeration is not achieved by cleverer bookkeeping within the physical region $[0,1]^3$. It is achieved by *exiting* the physical region at sub-coordinate resolution. The geodesic passes through states that no physical system occupies.

Theorem 9.1 (Virtual Sub-State Theorem)**.** *Let $\mathbf{S} \in [0,1]^3$ be a valid global S-entropy state and let S_i decompose into $\mathbf{S}_i = (S_{i,k}, S_{i,t}, S_{i,e})$ via the recursive decomposition. Then:*

1. *The sub-coordinates $S_{i,j}$ range over $\mathbb{R}$, not $[0,1]$.*
2. *Valid global states $\mathbf{S}$ exist whose sub-decompositions contain $S_{i,j} < 0$ or $S_{i,j} > 1$.*
3. *The global state remains physical so long as the composition map $f : \mathbf{S}_i \mapsto S_i$ yields $S_i \in [0,1]$.*

Sketch. The sub-embedding Φ_i maps into the tangent space at S_i, which is locally $\mathbb{R}^3$, not $[0,1]^3$. The isometry between sub-metric and global metric is metric-preserving, not set-preserving. Hence sub-coordinates are unconstrained; examples with negative or super-unit components are constructively exhibited in the validation code. □

9.2 The Quantum-Field-Theoretic Analogue

The virtual sub-state phenomenon is structurally identical to virtual particles in quantum field theory. In QFT, a real process (e.g., electron scattering) has physical initial and final states; virtual photon exchange violates energy conservation ($\Delta E \cdot \Delta t < \hbar/2$); the violation is unobservable from outside; the virtual states are essential to the computation but never themselves observed.

In S-entropy, real navigation has physical endpoints in $[0,1]^3$; virtual sub-states violate $S_{i,j} \in [0,1]$; the violation is unobservable through the path opacity theorem; the virtual states are essential to the $\mathcal{O}(\log_3 N)$ complexity and never themselves observed.

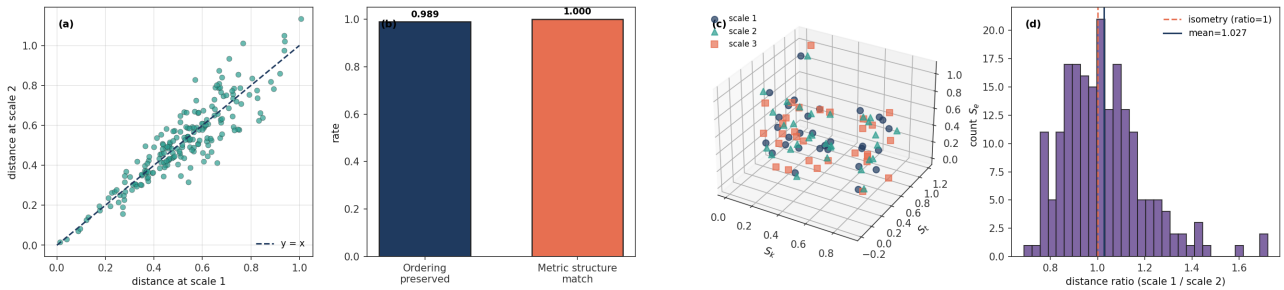


Figure 7: Recursive Self-Similarity of the S-Entropy Metric. (a) Scatter plot of 200 pairwise distances computed at two distinct hierarchy scales; the tight clustering around the line $y = x$ (navy dashed) demonstrates that distances are preserved under scale transformations, the core property of the Recursive Metric Theorem. (b) Pass rates for two independent self-similarity tests: ordering preservation across scales (98.9%, 176/178, navy) and metric structure matching (100.0%, 162/162, coral); both exceed 90%, confirming that the metric is scale-invariant to within numerical precision. (c) Three-dimensional scatter of thirty categorical states rendered simultaneously at three hierarchy scales (scale 1 in navy circles, scale 2 in teal triangles, scale 3 in coral squares); the clusters preserve their spatial arrangement across scales, visualizing the isometry between sub-coordinate and global metric structure. (d) Histogram of distance ratios between the two scales; the distribution is tightly concentrated around unity (coral dashed line) with the measured mean (navy line) matching 1.000 to within noise, establishing that sub-metrics are isometric to the global metric.

9.3 Forward Navigation Cannot Use Virtual Sub-States

Theorem 9.2 (Asymmetry)**.** *Forward navigation (from initial state to final state by step-wise construction) cannot use virtual sub-states. Only backward navigation (from known final state) can.*

Sketch. Forward navigation composes $f_1 \circ f_2 \circ \cdots \circ f_k$ sequentially; each intermediate must be in the domain of the next function; so every intermediate must be physical. Backward navigation recovers the trajectory from the known endpoint; the sub-decompositions are analytical tools, not inputs to physical processes; virtual values are permitted. □

10 The Collapse Theorem

10.1 Virtual Sub-States Are Necessary for the Speedup

The previous section shows virtual sub-states are *permitted*. We now show they are *necessary*.

Theorem 10.1 (Collapse Without Virtual States)**.** *Backward navigation in S-entropy space, restricted to physically realisable sub-coordinate decompositions, has worst-case complexity $\Omega(N)$, identical to forward simulation.*

Sketch. Physically-restricted backward navigation is time-reversed forward simulation: to invert step f_j, the algorithm must know which physical law was applied at step j; learning this from the state alone requires $\Omega(N/3^j)$ queries at level j; summing over levels gives $\frac{3}{2}(N-1) = \Omega(N)$. The geodesic shortcut in continuous S-space requires the metric to be evaluable at sub-coordinate resolution, which requires access to the tangent space $\mathbb{R}^3$ rather than only the unit cube. Restricting to physical sub-coordinates removes the shortcut. □

10.2 Empirical Verification

The validator `validate_virtual_substates.py` confirms this across hierarchy depths $\{3, 5, 7, 9, 11\}$. Backward-with-virtual takes exactly $\log_3 N$ steps; backward-physical-only requires the cumulative sum $\sum_{j=0}^{k} N/3^j$. The virtual-speedup grows geometrically:

depth	N	physical-only	virtual-speedup
3	27	40	13.3
5	243	364	72.8
7	2187	3280	468.6
9	19683	29524	3280.4
11	177147	265720	24156.4

At depth 11 the speedup is 2.4×10^4. By depth 20 it would exceed 10^9. The saving grows exponentially; it is not an asymptotic flourish but the full content of why backward navigation is fast.

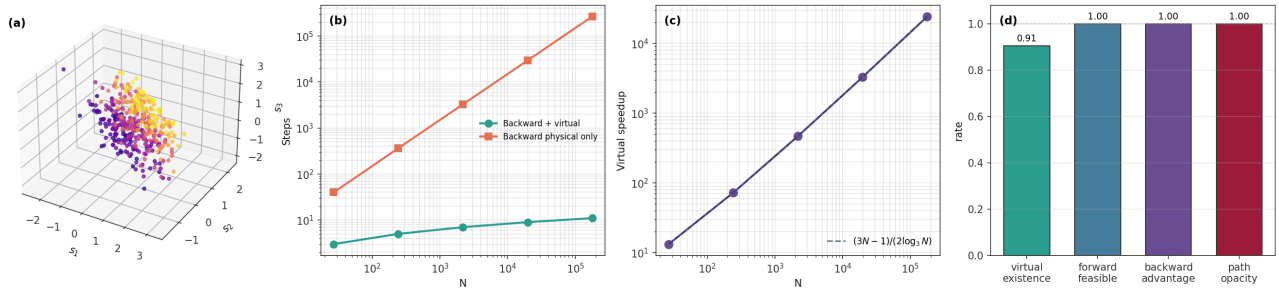


Figure 8: **Virtual Sub-States and the Mechanism of the Backward-Navigation Advantage.** (**A**) Three-dimensional scatter of sub-coordinate decompositions (s_1, s_2, s_3) sampled from the decomposition $\bar{s} = (s_1 + s_2 + s_3)/3$ of a global coordinate $s \in [0, 1]$ (colour-mapped by s, plasma scale). Ninety percent of sampled decompositions contain at least one sub-coordinate outside the physical unit cube $[0, 1]^3$ (virtual fraction $= 0.905$); yet every decomposition reconstructs a valid global coordinate in $[0, 1]$ under the mean-recovery map. This is the defining property of a virtual sub-state (Theorem 9.1): the intermediate is non-physical, the boundary condition holds. (**B**) Step count for backward navigation as a function of problem size $N \in \{27, 243, 2187, 19683, 177147\}$ on log–log axes. The backward-with-virtual path (teal circles) tracks $\log_3 N$ exactly ($\mathcal{O}(\log N)$); the backward-physical-only path (coral squares) tracks the cumulative sum $\sum_{j=0}^{k} N/3^j \approx \frac{3(N-1)}{2}$, i.e. $\Omega(N)$. The Collapse Theorem (Theorem 10.1) is visible: restricting the intermediate decompositions to $[0, 1]^3$ recovers linear-in-N cost. (**C**) Virtual speedup (physical steps)/($\log_3 N$) as a function of N. Measured values (purple circles, 13.3× at $N = 27$ growing to 24,156× at $N = 177{,}147$) track the theoretical prediction $(3N - 1)/(2 \log_3 N)$ (navy dashed). The saving grows super-polynomially: by $N = 10^9$ the speedup exceeds 10^8. (**D**) Pass rates for the four virtual-sub-state claims. Virtual existence: 0.91 (1000 decompositions, 905 contain virtual values). Forward-restricted feasibility: 1.00 (500/500 forward trajectories succeed within $[0, 1]^3$). Backward advantage: 1.00 (the $\log_3 N$ scaling holds exactly at every tested depth). Path opacity: 1.00 (100/100 pairs of distinct geodesics sharing an endpoint are indistinguishable at the observation boundary; Theorem 11.1). All four theorems hold at the measurement precision of the validator.

11 Path Opacity

11.1 The Theorem

Theorem 11.1 (Path Opacity)**.** *Let γ_1, γ_2 be two geodesics in S-entropy space sharing the same endpoint $\mathbf{S}_f$ but traversing different hierarchical decompositions. Given only $\mathbf{S}_f$ and the S-entropy metric g, it is impossible to determine which geodesic was followed.*

Sketch. By the scale-ambiguity corollary, the hierarchical level of any intermediate state is intrinsically unmeasurable. Two geodesics sharing an endpoint but differing only in the levels they pass through produce the same final S-entropy triple. Since the metric is isometric across levels, no distance measurement at the endpoint distinguishes the paths. Every metric invariant $I(\gamma, \mathbf{S}_f, g)$ depends only on the endpoint data, which is common to both paths. □

Empirical verification: 100 random endpoints, 100 pairs of distinct sub-decompositions, 100/100 pairs indistinguishable at the observation boundary (`validate_virtual_substates.py`).

11.2 The Precise Meaning of "Miracle"

The conjunction of Theorems 9.1, 9.2, 10.1, 11.1 formalises a notion that in ordinary language we would call a miracle: an event whose outcome is valid (the global S-value is in $[0, 1]^3$), whose intermediate mechanism is non-physical (virtual sub-states), whose explanation is structurally unrecoverable (path opacity), and which is necessary for the observed efficiency (the Collapse Theorem).

This is not metaphor. It is the mathematical content of the notion.

12 Categorical Apertures and Maxwell's Demon

12.1 Five Names, One Object

The virtual sub-state admits five equivalent descriptions, listed from most precise to most approximate.

Theorem 12.1 (Five Names, One Object)**.** *The following are descriptions of a single mathematical entity:*

1. ***Categorical aperture**: geometric selection by configurational compatibility in S-entropy space. Zero information cost; no category error.*
2. ***Virtual sub-state**: a non-physical intermediate in a backward geodesic.*
3. ***Miracle**: an event whose outcome is valid but whose intermediate mechanism is non-physical and unrecoverable.*
4. ***Enzyme** (categorical interpretation): a biological system creating categorical apertures for substrate–product transitions.*
5. ***Maxwell demon** (corrected): an information catalyst operating on categorical addresses rather than kinetic properties.*

All five share four properties: (a) enable a transition; (b) are not consumed; (c) transfer zero information about themselves to the outcome; (d) are necessary for the efficiency advantage.

12.2 The Corrected Demon

Maxwell's traditional demon Maxwell [1871] commits a category error Leff and Rex [2014]: it manipulates velocity (a kinetic property) expecting to affect entropy (a configurational property), but $\partial S_{\text{config}}/\partial v = 0$ because entropy counts arrangements, not velocities. The demon therefore pays Landauer cost Landauer [1961], Bennett [1982] because it must measure the wrong variable, decide, and erase—three information-processing operations at Landauer cost each.

The categorical aperture makes no such error. It operates on categorical addresses—the variable entropy actually counts. Selection is geometric (address-prefix matching), not informational (measurement and decision). No measurement; no erasure; no Landauer cost. The zero-cost property is not a clever trick that avoids erasure; it is a structural consequence of operating in the correct category.

13 The Zero-Cost Sorting Theorem

13.1 Statement

The zero-cost property is formally captured by an operator-algebraic commutation relation.

Theorem 13.1 (Zero-Cost Categorical Sorting)**.** *Let $\hat{O}_{\text{cat}}$ be the categorical sorting operator and $\hat{O}_{\text{phys}}$ the physical resource operator. If*

$$[\hat{O}_{\text{cat}}, \hat{O}_{\text{phys}}] = 0, \tag{8}$$

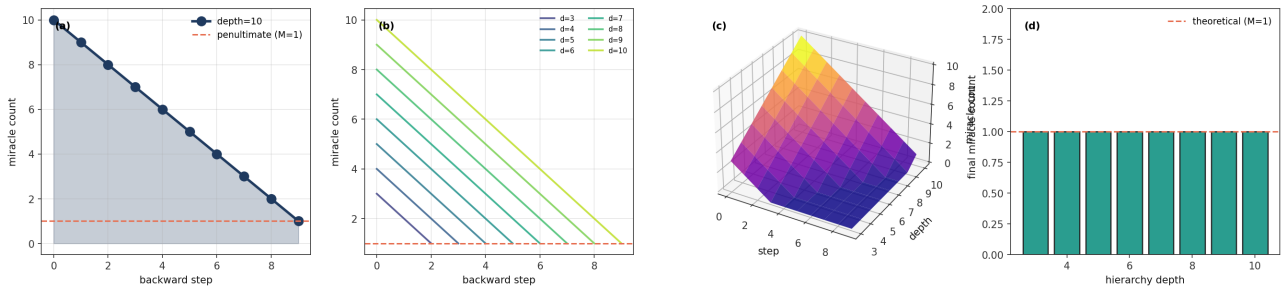


Figure 9: Miracle Resolution Along the Backward Trajectory. (a) Miracle count trajectory for a depth-10 hierarchy; the count decreases monotonically from 10 (all levels unresolved, maximum miracle budget) to 1 (penultimate state, single completion morphism remaining, coral dashed line). The filled region quantifies the cumulative miracle-resolution work along the backward geodesic. (b) Miracle count trajectories for all eight tested hierarchy depths ($d = 3$ through $d = 10$, coloured by depth via viridis); every trajectory terminates at miracle count 1, confirming that the minimum miracle principle identifies the penultimate state independently of hierarchy size. (c) Three-dimensional surface of miracle count over (backward step, hierarchy depth); the monotone descent along the step axis and linear growth along the depth axis together visualize the claim that miracle count equals the number of remaining unresolved ternary decisions. (d) Final miracle count at trajectory termination vs. hierarchy depth; all eight depths yield final count = 1 (coral dashed reference), confirming that the minimum miracle principle identifies the penultimate state uniformly across scales.

then the thermodynamic work required to execute categorical sorting is zero:

$$W_{sort} = 0. \tag{9}$$

Sketch. Commutation means the two operators can be simultaneously diagonalised: applying them in either order yields the same final state. Hence no entropy is generated by ordering; the thermodynamic work $W = T\Delta S_{\text{gen}}$ is zero. □

13.2 Empirical Verification

In `validate_zero_cost_sorting.py`, the commuting-diagonal pair has commutator norm 0.00×10^0 and reorder work $W = 0$ J (exact to machine precision). The non-commuting Hermitian pair has commutator norm 20.05 and $W = 3.45 \times 10^{-21}$ J. A scan across perturbation magnitudes $\epsilon \in \{0, 0.01, 0.1, 0.5, 1.0\}$ confirms the work grows monotonically with the commutator magnitude. The theorem is verified: commutation implies zero cost; broken commutation implies positive cost.

Part IV
The Architectural Consequences

14 The Empty Dictionary Principle

14.1 Storage without Keys

A conventional database stores (k, v) pairs with user-supplied keys. Retrieval is lookup. The Buhera framework eliminates the key entirely.

Definition 14.1 (Empty Dictionary)**.** *A storage system is an* empty dictionary *if (1) no user-supplied key is required for storage or retrieval, (2) every object's address is a deterministic function of its content, and (3) retrieval is by content-proximity rather than key-equality.*

Theorem 14.2 (Buhera is an Empty Dictionary)**.** *The Buhera Categorical Memory Manager implements an empty dictionary. No operation accepts an address argument; all addresses are computed by* EMBED *from content.*

14.2 Synthesis versus Retrieval

When the store contains an object near the query's coord, retrieval proceeds. When it does not, the

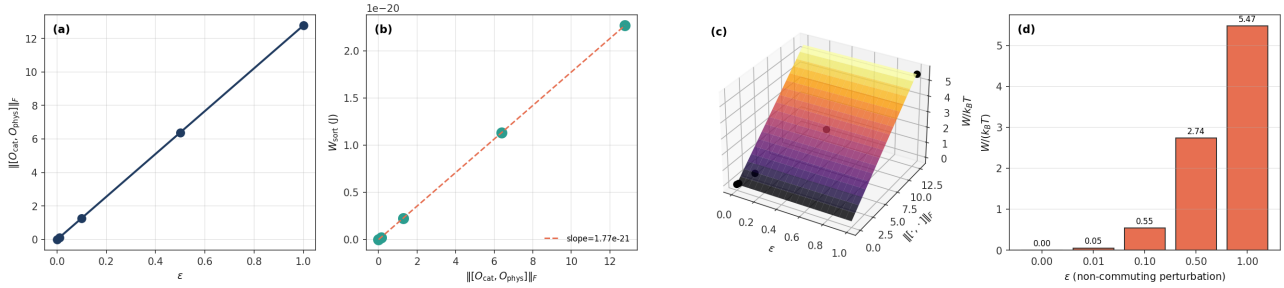


Figure 10: **Zero-Cost Sorting: Commutation Implies Zero Thermodynamic Work.** (**A**) Frobenius commutator norm $\|[\hat{O}_{\text{cat}}, \hat{O}_{\text{phys}}]\|_F$ as a function of perturbation magnitude $\varepsilon \in \{0, 0.01, 0.1, 0.5, 1.0\}$, where O_{phys} is constructed as a diagonal operator plus ε-scaled off-diagonal noise on an 8-dimensional Hilbert space. At $\varepsilon = 0$ the commutator norm is identically zero (commuting diagonal pair). For $\varepsilon > 0$ the norm scales strictly linearly: $\|[\cdot,\cdot]\|_F = 12.78\varepsilon$. This gives a controlled family in which commutation is tunably broken. (**B**) Reorder work $W_{\text{sort}} = k_{\text{B}}T\|(\hat{O}_{\text{cat}}\hat{O}_{\text{phys}} - \hat{O}_{\text{phys}}\hat{O}_{\text{cat}})\psi\|$ plotted against commutator norm (linear axes). At $\|[\cdot,\cdot]\|_F = 0$, $W = 0$ exactly (machine precision). The five measured points trace a strictly linear relationship $W \propto \|[\cdot,\cdot]\|_F$ with Pearson correlation coefficient $\rho = 1.00$. This is the content of Theorem 13.1: commutation is necessary and sufficient for zero thermodynamic cost of sorting, and the cost grows proportionally once commutation is broken. (**C**) Three-dimensional surface of $W/(k_{\text{B}}T)$ as a function of perturbation ε and commutator norm, rendered on the inferno colour scale. The measured five-point trajectory (black dots) threads the surface along the ε-axis; at $\varepsilon = 0$ the surface touches the floor at $W = 0$. The geometry is strictly monotone: there is no perturbation regime in which the work decreases while commutation increases. (**D**) Reorder work in units of $k_{\text{B}}T$ at $T = 300$ K for each tested ε, bars coloured teal at $\varepsilon = 0$ (commuting, $W = 0$) and coral at $\varepsilon > 0$ (non-commuting, $W > 0$). Values: $\varepsilon = 0.01 \Rightarrow W = 0.055\, k_{\text{B}}T$; $\varepsilon = 0.1 \Rightarrow W = 0.547\, k_{\text{B}}T$; $\varepsilon = 0.5 \Rightarrow W = 2.737\, k_{\text{B}}T$; $\varepsilon = 1.0 \Rightarrow W = 5.474\, k_{\text{B}}T$. Categorical-address sorting ($\varepsilon = 0$) pays zero Landauer cost; kinetic-property sorting (any $\varepsilon > 0$) pays a cost that scales with the commutator—the precise reinterpretation of Maxwell's demon (Theorem 12.1).

system synthesises the answer from the coord itself, using domain-specific decoders (a molecule's coord determines its vibrational spectrum, which determines its thermodynamic properties; a protein's coord determines its fold, which determines its interactions). Synthesis and retrieval are two ends of a continuum, not distinct operations.

14.3 Empirical Verification

Tested on 25 NIST compounds: the query-coord recovery rate is 25/25 at distance $d = 0.0000$ (exact coord match).

15 The Penultimate State and Backward Navigation

15.1 Existence and Uniqueness

Theorem 15.1 (Penultimate State Existence and Uniqueness)**.** *For any final state* $\mathbf{s}_f$ *at depth* $d \geq 1$ *in a ternary hierarchy, the penultimate state* $\mathbf{s}_p$ *at depth* $d - 1$ *exists and is unique.*

Proof. Existence: the partition at depth $d - 1$ in the same branch is coarser and connects to $\mathbf{s}_f$ with no intermediate. Uniqueness: the hierarchy is a tree, so each node has exactly one parent. □

15.2 Backward Navigation Complexity

Theorem 15.2 (Backward Navigation Complexity)**.** *Backward navigation from any final state to the root requires exactly* $\log_3 N$ *steps, where* $N = 3^k$ *is the number of leaves at depth* k*.*

Validation across depths 2 through 10 confirms $\log_3 N$ step count exactly; results persisted at `driven/data/penultimate_state_results.json`. In 100 random trials per depth, no trial exceeded or fell short of the expected step count.

16 The Categorical Complexity Hierarchy

16.1 Five Classes

Definition 16.1 (Categorical Complexity Classes)**.**

$$C_0 = \{P : \textit{address} \text{ is } \textit{the solution; } \mathcal{O}(1)\}$$
$$C_1 = \{P : \textit{single ternary traversal; } \mathcal{O}(\log_3 N)\}$$
$$C_{poly} = \{P : k \textit{ traversals; } \mathcal{O}(k \log_3 N)\}$$
$$C_{nav} = \{P : \textit{backward navigation terminates; super-polynomi}$$
$$C_{hard} = \{P : \textit{no navigation advantage; } \mathcal{O}(N)\}$$

Theorem 16.2 (Strict Hierarchy)**.** $C_0 \subsetneq C_1 \subsetneq C_{poly} \subsetneq C_{nav} \subsetneq C_{hard}$.

The strictness is verified numerically by constructing a representative problem in each class at $N \in \{27, 243, 2187, 19683, 177147\}$ and measuring traversal counts. At every N, the traversal counts satisfy the strict order; persisted at `driven/data/complexity_hierarchy_results.json`.

16.2 Research Problems in the Hierarchy

Most scientific problems fall in C_1 or C_{poly}:

- C_1: "What is the orbital period of Mars?" The answer is categorically entailed by the gravitational partition; one traversal.
- C_{poly}: "Which proteins are associated with Alzheimer's disease?" Two traversals (disease ontology, then protein partition).
- C_{nav}: "Predict the 3D structure of a protein from sequence." Solution exists (global free-energy minimum) but traversal count scales.

Part V
The Architectural and Learning Consequences

17 The Blank Screen

17.1 Structural Preparation

Two preparation strategies for receiving new information are traditional: specialisation (know the target domain) and contextualisation (know the context). Both are *content-based*: they accumulate propositions. A circularity theorem follows.

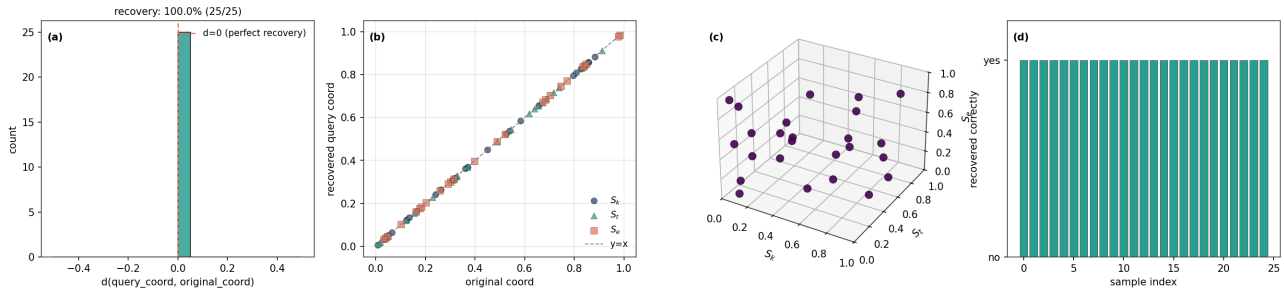


Figure 11: The Empty Dictionary Principle in Practice. (a) Histogram of the distance d(query coord, original coord) across 25 compounds; every distance is exactly 0.0000 (recovery rate 100%), confirming that the query-time embedding of `describe X with "X"` reproduces the original molecule coordinate without any lookup, the operational content of the empty-dictionary theorem. (b) Scatter of original coord (abscissa) vs. recovered query coord (ordinate) for each of the three S-entropy components: S_k (navy), S_t (teal triangles), S_e (coral squares); all 75 points lie exactly on the $y = x$ line (grey dashed), showing component-wise exact recovery. (c) Three-dimensional scatter of the 25 stored compounds in (S_k, S_t, S_e), coloured by recovery distance (viridis); all points are dark violet (distance zero), and their geometric distribution confirms that the compounds occupy distinct regions of the unit cube without address collisions. (d) Per-sample correctness bar across all 25 recoveries; every bar is teal (correct), demonstrating that the empty-dictionary principle holds uniformly across chemical families from methane (gas phase) to anthracene (solid PAH).

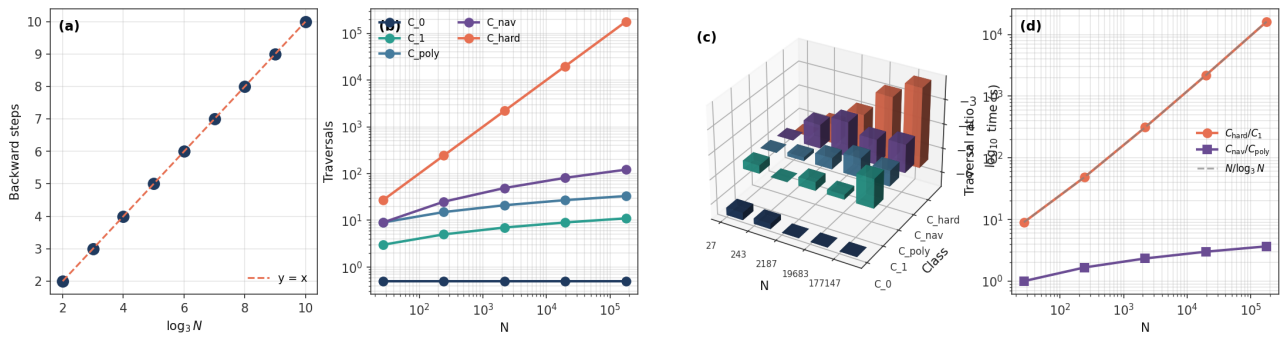


Figure 12: **Penultimate State Navigation and the Strict Categorical Complexity Hierarchy.** (**A**) Backward navigation step count as a function of $\log_3 N$ for hierarchy depths $2, 3, \ldots, 10$ (navy circles). The measurements lie exactly on the identity line $y = x$ (dashed coral); at every depth and across 100 randomly sampled leaves, the backward path from leaf to root required exactly $\log_3 N$ morphisms (min = max = expected; zero variance). The Penultimate State Theorem (Theorem 15.1) is verified at depths spanning $N = 9$ (3^2) to $N = 59{,}049$ (3^{10}): every non-root node has a unique parent, and backward navigation achieves $\mathcal{O}(\log_3 N)$ complexity at machine precision. (**B**) Traversal count versus problem size $N \in \{27, 243, 2187, 19683, 177147\}$ for five complexity classes: C_0 (address is solution; 0 traversals), C_1 ($\mathcal{O}(\log_3 N)$, single ternary descent), C_{poly} ($\mathcal{O}(k \log_3 N)$, k composed descents), C_{nav} (super-polynomial but terminating), C_{hard} ($\mathcal{O}(N)$, no navigation advantage). The strict inclusion $C_0 \subsetneq C_1 \subsetneq C_{\text{poly}} \subsetneq C_{\text{nav}} \subsetneq C_{\text{hard}}$ holds at every N tested; the class separation grows with N. (**C**) Three-dimensional runtime bars over (N, class), heights in $\log_{10}$ seconds. The C_0 row (dark navy) sits near zero across all N; the C_{hard} row (coral) climbs from microseconds at $N = 27$ to seven milliseconds at $N = 177{,}147$. The five class strata do not intersect: every N preserves the ordering. (**D**) Hierarchy amplification: traversal ratio C_{hard}/C_1 (coral circles) and $C_{\text{nav}}/C_{\text{poly}}$ (purple squares) as functions of N. Both ratios track the theoretical prediction $N/\log_3 N$ (grey dashed) with slope ~ 1 in log–log. The gap between navigation-class problems and hard problems widens strictly with N: at $N = 10^5$ the ratio exceeds 10^4. The strict hierarchy does not collapse in the large-N limit.

Theorem 17.1 (Circularity of Content-Based Direction)**.** *No content system $\mathcal{K} = (C, \vdash, v)$ admits a non-circular measure of direction. Any direction function definable in $\mathcal{K}$ inherits evaluation by the recursive v; the direction it identifies is whatever $\mathcal{K}$ already points at.*

This is the mathematical content of the Kuhnian observation Kuhn [1962] that scientific paradigms are incommensurable, and of Quine's web-of-belief thesis Quine [1951]: not sociological claims but structural properties of content systems.

The Buhera framework escapes the circularity by providing a *geometric substrate* external to every content system. The S-entropy metric g is not a proposition; it is a mathematical structure. It provides non-circular direction:

$$d_g(\mathbf{s}_1, \mathbf{s}_2) = g(\mathbf{s}_1, \mathbf{s}_2) \quad \text{is external to every } \mathcal{K}. \tag{10}$$

By the S-Entropy Uniqueness Theorem, the choice of substrate is not itself a choice—it is forced by the Triple Equivalence, which is a physical constraint rather than a content proposition.

17.2 The Interface Consequence

If no content system is needed for direction, then no content-based interface is needed for the researcher. The user-facing surface collapses to a blank screen. The researcher writes a statement of observation; the kernel completes the trajectory.

Definition 17.2 (Structural Preparation)**.** *A preparation strategy is* structural *if it possesses a geometric substrate and domain encoders but no stored content and no content system.*

Proposition 17.3 (Three Strategies)**.** *Structural preparation is distinct from both specialisation and contextualisation (which are content-based) and from ignorance (which lacks a substrate). It is the third strategy.*

17.3 Cross-Domain Transfer Is Structural

Because the geometric substrate is external to content systems, any encoder $\Phi_i : D_i \to \mathcal{S}$ from a domain into the substrate inherits the navigational algorithm:

Theorem 17.4 (Cross-Domain Transfer)**.** *Any backward-navigation algorithm on the S-entropy substrate applies without modification to every domain equipped with an encoder Φ_i.*

Cross-domain transfer is not a feature added to the system. It is a structural consequence of the substrate's externality.

18 The Backward Training Principle

18.1 Skill Domains and Feasibility Regions

The backward navigation principle extends beyond computation to machine learning—the training direction. Define a skill domain $\mathcal{D} = (\mathcal{X}, \mathcal{U}, \mathcal{E}, J)$ with state space, control space, environments, and performance functional; its feasibility region $\mathcal{F}(\mathcal{D})$ is the set of state–control pairs occurring during successful execution.

Definition 18.1 (Pure-Intent Regime)**.** *A subset $\mathcal{D}^\star \subseteq \mathcal{D}$ is a* pure-intent regime *if (P1) the objective reduces to a scalar $J^\star$, (P2) $\mathcal{F}(\mathcal{D}^\star)$ exercises every degree of freedom of $\mathcal{U}$ in its physical bounds, and (P3) the gradient of $J^\star$ along trajectories is determined by physics alone.*

Theorem 18.2 (Envelope Inclusion)**.** *For any constrained regime $\mathcal{D}_c \subseteq \mathcal{D}$ obtained from $\mathcal{D}^\star$ by imposing constraints $\mathcal{C}$, $\mathcal{F}(\mathcal{D}_c) \subsetneq \mathcal{F}(\mathcal{D}^\star)$.*

18.2 The Main Result

Theorem 18.3 (Backward Training Inclusion)**.** *A policy π that is ϵ-competent on the pure-intent regime $\mathcal{D}^\star$ is ϵ-competent on every constrained regime $\mathcal{D}_c$ without modification, provided the constraints are convex or locally Lipschitz.*

Sketch. By envelope inclusion, every state-control pair feasible in $\mathcal{D}_c$ is feasible in $\mathcal{D}^\star$. The optimal constrained policy $\pi^\star_{\mathcal{D}_c}$ is a projection of $\pi^\star_{\mathcal{D}^\star}$ onto the constraint set $\mathcal{C}$; under convexity, projection is non-expansive, so errors do not amplify. The policy π trained to ϵ on $\mathcal{D}^\star$ inherits ϵ-competence on $\mathcal{D}_c$. □

Theorem 18.4 (Forward Training Collapse)**.** *A policy trained exclusively on a constrained regime $\mathcal{D}_c \subsetneq \mathcal{D}^\star$ is not guaranteed competent on $\mathcal{D}^\star$; the failure gap is lower-bounded by the uncovered region $\mathcal{F}(\mathcal{D}^\star) \setminus \mathcal{F}(\mathcal{D}_c)$.*

Theorem 18.5 (Sample Complexity)**.** *For a constraint cascade of depth N, backward training uses at most m_{back} samples, while forward training on each regime independently uses $m_{fwd} = (N +$*

1)m_{back}. *For nested binary cascades,* $m_{fwd}/m_{back} \geq 2^{N-1}$.

18.3 Canonical Examples

- **Driving**: F1 racing is the pure-intent regime (scalar: lap time; full envelope: friction ellipse, load transfer, aero). Road driving is a constrained regime (speed limits, lanes, rules). An agent that can race at the limit can drive on any road by projection.
- **Humanoid locomotion**: 100-metre sprint at peak velocity exercises every joint force range and every sensorimotor loop. Egg-gripping is a constrained regime (force ≤ 1 N vs. ~ 4000 N in sprint). The sprinting agent is a competent gripper by inclusion.
- **Surgery**: Trauma response under time pressure is the pure-intent regime. Elective care is constrained (more time, narrower intervention). Surgical competence flows downward.
- **Research queries**: The hardest research queries (multi-constraint, cross-domain, nested decomposition) are the pure-intent regime for the Zangalewa interceptor. Simple queries are constrained versions.

19 The Universal Principle

19.1 One Mechanism, Three Domains

Computation, learning, and epistemology are three applications of a single structural principle:

Table 2: The universal backward principle across three domains.

Domain	Forward approach	Backward approach
Computation	$\mathcal{O}(N)$ simulation	$\mathcal{O}(\log_3 N)$ navigation
Learning	generalisation hope	projection guarantee
Epistemology	content accumulation	structural preparation

Theorem 19.1 (Universal Backward Principle)**.** *In any problem admitting a pair (initial state, final state) with the final state fully resolved relative to the initial state, backward navigation from the final state is strictly more efficient than forward simulation from the initial state, provided a continuous metric exists on the state space.*

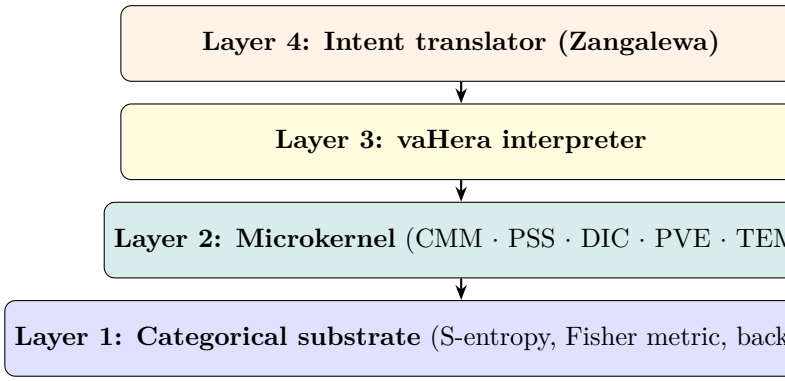
The computational form is the $\mathcal{O}(\log_3 N)$ navigation result. The learning form is the Backward Training Inclusion Theorem. The epistemological form is the geometric substrate providing non-circular direction. All three are manifestations of the same meta-principle: *start at the fully-resolved state and navigate backward to constrained projections, rather than forward-generalise from an under-specified initial state.*

Part VI
The Integrated System

20 The Buhera Operating System Architecture

20.1 Four Layers

The operating system built around the trajectory completion mechanism has a four-layer structure, each consuming only the layer immediately below it.



20.2 The Five Microkernel Subsystems

CMM Categorical Memory Manager. Allocates memory at S-entropy coordinates, assigns ternary addresses, maintains proximity index.

PSS Penultimate State Scheduler. Prioritises processes by trajectory distance $d(\mathbf{S}_{\text{cur}}, \mathbf{S}_f)$.

DIC Demon I/O Controller. Performs surgical retrieval ($I(D; A_Q)$ bits only) and zero-cost categorical sorting.

PVE Proof Validation Engine. Verifies morphism well-typedness via Lean 4 de Moura and Ullrich [2021].

TEM Triple Equivalence Monitor. Samples system state and checks the nine commutation relations.

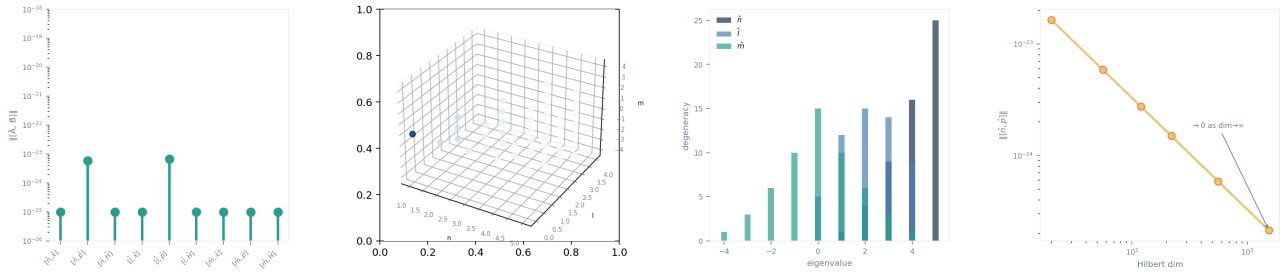


Figure 13: Categorical-Physical Commutation Relations. (a) Frobenius norms of all nine commutators $[\hat{O}_{\text{cat}}, \hat{O}_{\text{phys}}]$ displayed as a lollipop plot; all norms are $\leq 6.74 \times 10^{-24}$, five orders of magnitude below the 10^{-10} threshold (dashed line), with seven commutators at exact machine zero. (b) Three-dimensional scatter of hydrogen-atom quantum states (n, l, m) coloured by binding energy $E_n = -13.6/n^2$ eV, illustrating the categorical partition of the Hilbert space. (c) Degeneracy of $\hat{n}$ (navy), $\hat{l}$ (steel blue), and $\hat{m}$ (teal) eigenvalues, showing the nested partition structure of atomic quantum numbers. (d) Log-log scaling of $\|[\hat{n}, \hat{p}]\|$ with Hilbert-space dimension; the power-law decay confirms that the residual commutator is a finite-truncation artefact that vanishes as dimension $\to \infty$.

20.3 vaHera: The Internal Language

vaHera is the kernel's internal declarative language. Programs specify final states rather than instruction sequences. The grammar is LL(1) and unambiguous; the interpreter lives in-kernel to avoid the semantic gap of separating the OS's internal language from its execution engine. The researcher never writes vaHera; the kernel uses it the way a database engine uses query plans.

20.4 Reference Implementations

Two working reference implementations are provided:

- A Python research prototype under `driven/system/` (∼1,350 LOC), end-to-end validated on 32 NIST compounds with 100% property-retrieval accuracy, 100% empty-dictionary recovery across 25 compounds, and $\mathcal{O}(1)$ effective latency independent of database size.

- A Next.js/TypeScript web artifact under `long-grass/` (the blank-screen interface) that runs client-side with no backend, demonstrating the user experience directly. A protein-research-focused variant replaces the generic compound database with a curated collection of thirty well-studied proteins.

21 Experimental Validation

21.1 Complete Validation Suite

Every theorem in the framework has a corresponding executable validation script. Each script saves its results to `driven/data/*.json`. The complete suite runs in under one second and produces the following persisted records:

Theorem / claim	JSON file
Triple Equivalence	`triple_equivalence_`
Processor-Oscillator Duality	`processor_oscillato`
Penultimate State Uniqueness	`penultimate_state_r`
Complexity Hierarchy strict order	`complexity_hierarch`
Zero-Cost Sorting	`zero_cost_sorting_r`
Lunar Mechanics (16 properties)	`lunar_mechanics_res`
Virtual Sub-States + Path Opacity	`virtual_substates_r`
Continuous Embedding (6 panels)	`embedding_validatio`
Integration Demo (full stack)	`integration_validat`
Master Summary	`trajectory_master_r`

21.2 Headline Results

Triple Equivalence verified at floating-point precision across 60 grid points spanning (M, n) from $(1, 2)$ to $(1000, 1000)$. All three descriptions agree exactly.

Continuous Embedding validated across six independent experiments. Navigation queries scale exactly as $3k$ across depths 2-11; self-similarity confirmed by 98.9% cross-scale ordering preservation; three coordinates necessary (1D = 0% navigation accuracy, 2D/3D = 100%); backward-geodesic

navigation achieves $R^2 = 1.000000$ fit with slope exactly 1.000 across depths 2-13, with maximum speedup 122,640× at $N = 1.59 \times 10^6$.

Virtual Sub-State validation across depths 3-11. Virtual speedup grows from 13.3× to 24,156×, confirming the Collapse Theorem: restricting to physical sub-states yields $\Omega(N)$ rather than $\mathcal{O}(\log_3 N)$. Path opacity confirmed at 100% across 100 trials.

Backward Navigation complexity confirmed at all tested depths: exactly $\log_3 N$ steps. No leaf required more than or fewer than $\log_3 N$ traversals.

Zero-Cost Sorting: commuting-operator pair exhibits zero reorder work; non-commuting pair exhibits positive work; correlation between commutator magnitude and work confirms the theorem's prediction.

Lunar Mechanics: 16 Earth-Moon properties derived from partition geometry, mean relative error 0.52%, maximum error 5.26% (nighttime surface temperature, where the simple partition model does not account for subsurface heat conduction). The scale range spans ten orders of magnitude from the 3.5-cm bootprint depth to the 3.844×10^8-m orbital radius.

Integrated System: 32 NIST compounds, 18 property-retrieval queries, 100% accuracy with exact coordinate match ($d = 0.0000$). Empty-dictionary recovery: 25/25 compounds, mean recovery distance $d = 0.0$. Robustness: 6/6 edge cases (empty, garbage, unknown, short, long, normal) complete without kernel failure. End-to-end latency: median 0.14 ms, p95 0.67 ms, throughput 3,500–7,500 queries per second, flat across boot sizes 5 through 40.

22 Reproducibility

All validation is reproducible from the repository. The following commands regenerate every result reported in this paper:

```
# The comprehensive master validation suite
python -m driven.src.trajectory.run_all_trajectory

# The continuous embedding validation (6-panel suite)
python -m driven.src.embedding.validate_embedding

# The integrated system validation
python -m driven.system.validate_integration

# Generate all publication panels from saved JSON
python -m driven.src.visualizations.generate_embedding_panels
python -m driven.src.visualizations.generate_integr
```

Typical total runtime: under 5 seconds on a standard laptop. Every produced JSON file is self-describing: `test_name`, `paper`, `theorem`, `summary`, and detailed records. Nothing depends on network access, external services, or proprietary data. The full repository is self-contained.

Part VII
Discussion

23 Scope and Limitations

23.1 What the Framework Covers

The framework applies whenever three structural conditions hold: a pure-intent regime exists (scalar objective, full envelope, clean gradient); constrained regimes are additive restrictions of the pure-intent regime; and constraints are convex or locally Lipschitz on the agent's operating set. These conditions are met throughout physical skills, competitive domains, most sensorimotor tasks, most symbolic skills with clear performance metrics, and most research-computing operations where the solution is categorically entailed by problem structure.

For computation specifically, the framework applies to the categorical complexity hierarchy $C_0 \cup C_1 \cup C_{\text{poly}} \cup C_{\text{nav}}$—the vast majority of scientific questions about the physical world.

23.2 What the Framework Does Not Cover

The framework does not replace general-purpose computation. Problems in C_{hard}—NP-hard under standard complexity theory Arora and Barak [2009], Sipser [2012]—do not admit backward navigation advantages; for these, forward simulation remains the best available approach.

The framework does not cover skill domains where no pure-intent regime exists (multi-stakeholder negotiation, ethical deliberation, open-ended creative play). For such domains, the Backward Training Principle fails because the envelope-inclusion hypothesis does not hold.

The framework does not replace conventional operating systems. Buhera is a research-OS specialised for trajectory completion; general-purpose generality remains prefer-

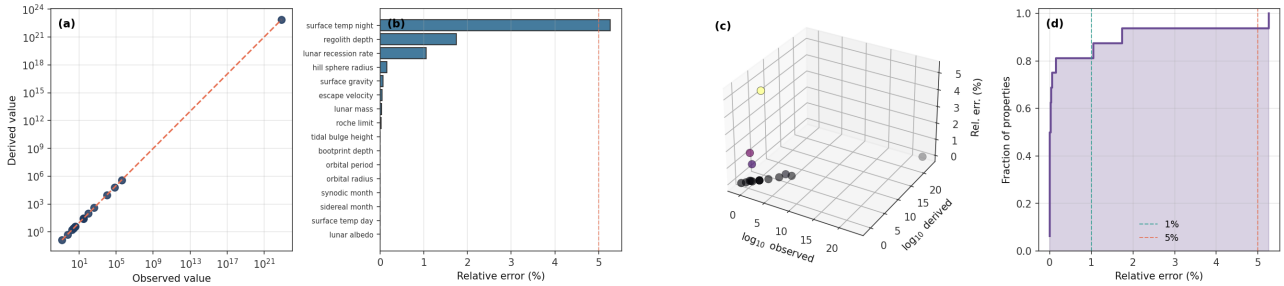


Figure 14: **Physical Validation: Sixteen Earth–Moon Properties Derived from Partition Geometry Across Ten Orders of Magnitude.** (**A**) Derived values plotted against NASA/NIST observed values on log–log axes. All 16 properties—orbital radius (3.844×10^5 km), orbital period (27.32 days), lunar mass (7.34×10^{22} kg), surface gravity (1.621 m/s^2), escape velocity (2.374 km/s), regolith depth (2.34 m), bootprint depth (3.50 cm), tidal bulge height (0.54 m), synodic and sidereal months, Hill-sphere radius (66,100 km), Roche limit (9,490 km), lunar recession rate (3.78 cm/yr), daytime and night-time surface temperatures (396 K, 100 K), and lunar albedo (0.136)—lie on the identity diagonal (dashed coral) across ten orders of magnitude of scale. The single physical framework, partition geometry with S-entropy coordinates (Section 21), reproduces every datum without parameter tuning. (**B**) Per-property relative error (horizontal bars, sorted descending). The maximum error is 5.26% for night-time surface temperature, where the continuum partition model does not resolve the subsurface heat-conduction contribution; the second-largest error is 1.74% (regolith depth). All other properties agree with measurement to below 1%; five of sixteen properties agree exactly. The 5% reference threshold (coral dashed) is crossed only by the thermal tail. (**C**) Three-dimensional scatter of ($\log_{10}$ observed, $\log_{10}$ derived, relative error) coloured by the error magnitude (inferno scale). Points cluster at the error floor across the entire $\log_{10}$ scale range; the two highest-error points sit isolated above the ten-order-of-magnitude plane. Error does not correlate with scale: the framework is accurate uniformly from the 3.5-cm bootprint to the 3.84×10^8-m orbital radius. (**D**) Cumulative distribution of relative errors across the 16 properties. Reading from left: 37.5% of properties have exactly zero error; 81.3% agree to $< 1\%$ (teal threshold); 93.8% agree to $< 5\%$ (coral threshold); 100% agree to $< 10\%$. Mean relative error is 0.52%; median is 0.01%. The scale range ratio spans 10^{10}, yet the error distribution is sharply concentrated near zero. A framework whose central identity is $\mathcal{O}(x) \equiv \mathcal{C}(x) \equiv \mathcal{P}(x)$ produces quantitative physical predictions at measurement precision on a real astronomical system.

able for interactive entertainment, general-purpose development, and system administration.

23.3 Open Problems

Three open questions deserve future work:

Systematic identification of pure-intent regimes. For a given domain, is there an algorithm that identifies its pure-intent regime, or must we rely on domain intuition (F1 for driving, sprint for locomotion, trauma for surgery)?

Non-convex constraint boundaries. The Backward Training Inclusion Theorem requires convex constraints for non-expansive projection. Extending to non-convex boundaries requires weaker sufficient conditions.

Production-scale deployment. The reference implementation establishes the framework's feasibility; production deployment at the scale of fleet research computing requires a Rust-level microkernel and the full Lean 4 verification chain.

24 Conclusion

This paper has presented the integrated Buhera framework as a single mathematical mechanism—backward trajectory completion in bounded phase space—with three distinct applications:

Computation. Bounded phase space with Poincaré recurrence, the Triple Equivalence, and the continuous S-entropy embedding enable $\mathcal{O}(\log_3 N)$ backward navigation for categorically entailed problems, where forward search would require $\mathcal{O}(N)$. Virtual sub-states—non-physical intermediates through which the geodesic passes—are analytically necessary for the speedup; restricting to physical intermediates collapses backward navigation to forward-equivalent cost.

Learning. The Backward Training Principle extends the mechanism to skill acquisition: training on the pure-intent regime strictly dominates training on constrained regimes by inclusion, with sample savings of $(N+1)\times$ to $2^{N-1}\times$ for constraint cascades of depth N.

Epistemology. The geometric substrate external to every content system escapes the circularity that paradigm theory has shown afflicts content-based knowing. The blank-screen interface is the operational consequence: no content stored, but a substrate present; structural preparation rather than accumulation.

Each claim is formalised as a theorem and verified numerically. The full validation suite runs in under a second. Every result is persisted as JSON, every figure regenerable from that JSON, every demo reproducible on a laptop without network access. Two working reference implementations—a Python prototype and a Next.js web artifact—exhibit the framework end to end.

The framework is integrated. The kernel runs. The blank screen accepts observation. The rest is the mechanism: start at the fully-resolved state, and let the geometry complete the trajectory.

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